Kuku Promo Finder: Find the Promos (Problem 1)

Author: Rakshit · **Repo:** `run.py` (engine), `app/` (Next.js viewer) · **Run:** `outputs/promos.meta.json`

1. PROBLEM, PRODUCT FRAMING & SCOPE

An editor watches every episode to pick 20-60s promos by hand, and new shows ship weekly. The tool reads timestamped screenplays and returns ranked promos: episode, every segment's start/end, hook, cliffhanger, and why it converts.

- **Conversion Framing:** A promo exists to create a curiosity gap strong enough that a viewer installs the app and pays to see the next moment. It does not summarize the plot and does not protect against spoilers; late-episode reveals are valid picks. Priorities: stakes legible within 3s, an immediate scroll-stopping hook, and unresolved payoff withheld at the cut. "Understood cold" is a secondary sanity check.
- **Assumptions & Caveats:** Screenplay text is the only input (no video), so visual judgements rely strictly on stage directions. Promos are single-episode cuts (composites remain experimental and disabled). *Conversion priority is an editorial assumption based on observed microdrama ad mechanics; there is no live conversion data.*

2. APPROACH: CODE ENFORCES CONSTRAINTS, AI JUDGES DRAMA

Step	Owner	Why
Parse 50 episodes into 3,727 timed beats	Code (`src/parser.py`)	Timestamps must be mathematically exact
Window discovery, tag hook/cliffhanger, score drama, conversion intent & payoff location	LLM (`gpt-6-luna` , one call per episode)	Identifies dramatic tension, emotional betrayal, and curiosity gaps
Hard constraints: 20-60s, hook starts ≤3s, cuts on real beat edges (snap ≤0.5s), verbatim quotes, cliffhanger in final 6s, payoff beat verification	Code (`src/validation.py`)	Models hallucinate timestamps and quotes. Cuts within 0.5s snap to spoken edges; others drop. Hallucinated payoff locations are discarded.
Dual-Judge adversarial audit panel (`gpt-6-luna` + `gemini-3.5-flash-lite`): inspects clip dialogue, camera beats, and taxonomy definitions	LLM panel (`src/stages.py` , `src/auditor.py`)	Catches resolution leaks, context deficits, visual dependency, and tag mismatches
Tiered Rank: consensus tiers (Tier 1 > Tier 2 > Tier 3) + editorial weights + diversity penalty	Code (`src/constants.py` , `src/ranker.py`)	Tier 1 (both judges pass) ranks first, then Tier 2 (split verdict: "needs editor look"). Repeat penalties prevent feed monotony.

Scoring Weights (editorial priors, uncalibrated): `hook 0.30` , `cliffhanger 0.25` , `payoff_withheld 0.20` , `injustice_tension 0.15` , `standalone_clarity 0.10` , `conversion_intent 0.00` .

3. RESULTS & MEASURED ABLATIONS (`OUTPUTS/ABLATIONS/`)

- **Headline Run (50 episodes):** 62 candidates proposed, 62 valid, 60 deduped across the season.
- **Independent Dual-Judge Panel (`gpt-6-luna` + `gemini-3.5-flash-lite`):**
  - Evaluated on all 60 candidates: Judge 1 pass rate **55.0%** (33/60), Judge 2 pass rate **63.3%** (38/60).
  - Raw agreement rate is **68.3%** (41/60), with Cohen's kappa **κ = 0.3493** (fair consensus).
  - Consensus audit tiers: **Tier 1 = 26, Tier 2 = 19, Tier 3 = 15** (0 audit errors). All 15 headline promos are Tier 1 consensus picks.
  - Gemini failed 3 previous top-15 promos: `ep28_cand_2` (`resolution_leak` : reveals execution & herbarium book evidence), `ep46_cand_1` (`resolution_leak` : High Priest dialogue resolves core mystery), and `ep34_cand_1` (`slow_burn` : courtroom preamble takes 15s to establish stakes).
- **Cliffhanger Trim-and-Revalidate:** When a cliffhanger beat ends before the final 6s tail window, code trims the cut to the beat's real end edge and re-verifies all hard constraints. In Ep 15, `ep15_cand_1` was recovered by trimming 7.5s (41.2s -> 33.7s).
- **Empty Episode Diagnosis:** 6 empty episodes remain in the main pool (`[6, 8, 10, 11, 17, 25]`). Episodes 6, 8, 10, 17 returned 0 model proposals (genuinely quiet dialogue); Episode 11 had cliffhanger quotes ending after cut points; Episode 25 had unsnapped start cuts. Episodes 3, 14, 22, 24, 38, 44, 47, 49 now yield valid candidates.
- **Payoff Location Verification:** 5 of top 15 promos have a code-verified payoff_location in the screenplay (Rank 2 Ep 2 `00:37.50` , Rank 3 Ep 3 `01:24.00` , Rank 13 Ep 35 `02:25.50` , Rank 14 Ep 18 `01:51.50` , Rank 15 Ep 1 `01:10.60`). The other 10 are `unknown` because the conflict resolves in later episodes or model-proposed timestamps did not snap to real beat edges (preventing hallucinated timestamps).

4. FAILURE MODES, TAXONOMY FIT & SYSTEM LIMITS

- **Invalid Offline Audit Replaced:** An earlier keyword audit passed 87/87 candidates due to unreachable score thresholds (`outputs/smoke_test_heuristic/promos.json` , commit `809d946`). Fixed: audits are labelled by method (`llm` / `rules` / `error`), rule lint is distinguished from quality, and single-judge runs refuse same-model self-auditing unless `--allow-self-audit` is explicit.
- **Taxonomy Mismatch:** In the top 15, **10 of top 15 hooks and 13 of top 15 cliffhangers** carry a model-proposed `new:` tag (e.g. `new:threatened_last_keepsake`). Novel tags receive default weight (0.90), so fine-grained taxonomy weights barely differentiate ranking without taxonomy consolidation.
- **Why Composites are Disabled:** Naive pairing across episodes severed character and emotional continuity. Safe composites require hook-first cuts followed by setup scenes with identical character IDs, validated by blind cold-read scoring.
- **Ground Truth & Recall:** Without a human-labeled ground truth (`eval/ground_truth.json`), absolute recall and precision remain **unmeasured**.

5. HUMAN IN THE LOOP & EDITOR HANDOFF

Editors review picks via the Next.js viewer or `outputs/PROMOS.md` , displaying audit tier badges, dual-judge critique breakdowns, flaw types, and cliffhanger trim indicators. `--export-csv` outputs `outputs/promos.csv` with one row per segment formatted in `mm:ss.ss` with verbatim quotes and verified `payoff_location` .

6. COST & EXECUTION PROFILE (MEASURED)

- **Committed Headline Run:** Extraction and audit panel executed with caching across OpenAI `gpt-6-luna` and Gemini `gemini-3.5-flash-lite`. Cached cost is **$0.0190**, with **$0.0002** live cost from discovery probes. Pure re-ranking is **$0.0000** and runs in **<0.1s**.
- **Stage Modularity:** Rerank, cold-read, and panel stages run independently with full token and cost provenance recorded in meta files.

7. TOOLS USED

Antigravity and Claude Code were used for pipeline implementation, validation harness, and UI. Runtime LLMs are OpenAI `gpt-6-luna` (extraction) and Google Gemini `gemini-3.5-flash-lite` (independent audit judge). See `AGENTS.md` for AI failure log.